

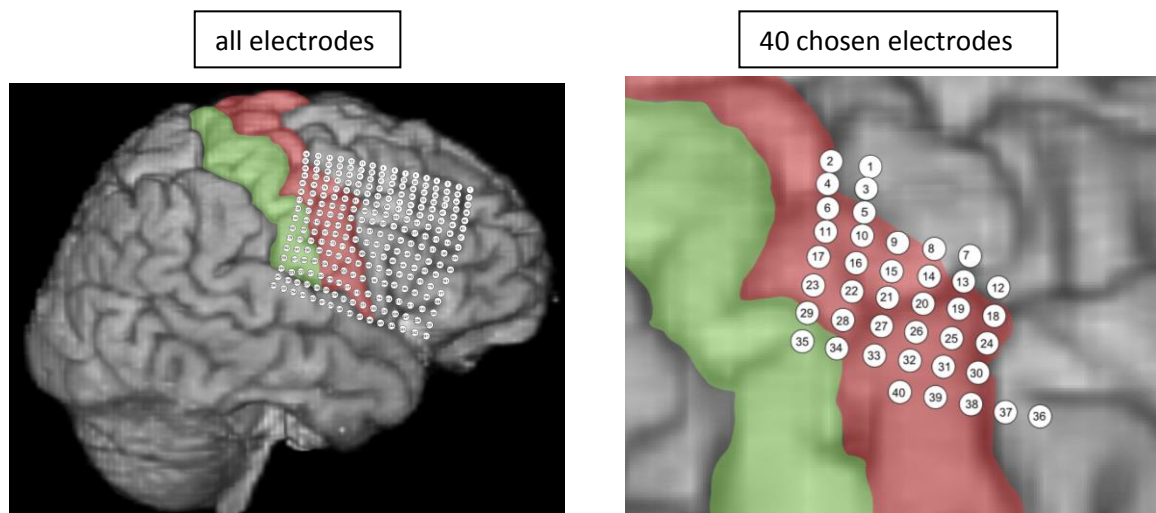
## Experimental paradigm and data structure

Most movement related Brain-Machine-Interface (BMI) research in humans has focused on decoding two or three classes of rather elementary real or imagined movements such as sticking out the tongue or squeezing the hand. However, to be able to control more complex devices with more degrees of freedom one would like to be able to discriminate more movement classes. In humans, gestures of the hand and the arm are increasingly used for computer interfacing. However, little is currently known about the feasibility of using different complex gestures to control BMIs. The aim of this study was to investigate whether subdural ECoG-recordings allow for reliable single trial multi-class discrimination between different gestures as well as different phases within gestures.

### Subjects

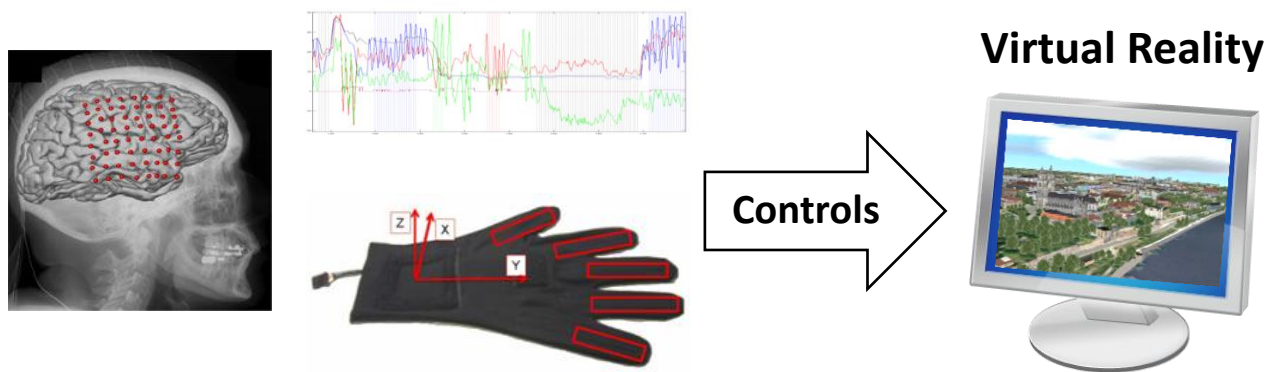
Five patients (S1 – S5) with subdurally placed electrode grids for localization of epileptic foci participated in this study. The location of the grid placement was determined only on medical considerations. Center-to-center electrode spacing was either 1 cm (8x8 electrodes) or 0.4 cm (16x16 electrodes).

This course will focus on the data set from **one subject** only. The data set from this subject originally includes 256 electrodes with an interelectrode spacing of 0.4 cm. In this course however, we will focus on only **40 of these electrodes**, in order to reduce the amount of data.



### Task

The patients' task was to use gestures of one hand to navigate through a virtual reality (VR) environment to collect tokens. Gestures were assessed with a data glove equipped with one bend sensor for each finger and a three-axis accelerometer. They performed five different gestures to navigate through the VR-environment: Index finger wiggle, waving the hand up and down with extended fingers, turning the hand palm up and down with extended fingers, turning the fist and waving the fist up and down. Based on the glove sensor data we manually assigned time intervals to gestures.



| Name              | Gesture                                                 | Performance    |
|-------------------|---------------------------------------------------------|----------------|
| <b>Finger</b>     | Flexion and extension of index finger                   | Moving forward |
| <b>Pitch hand</b> | Waving the hand up and down with extended fingers       | Left turn      |
| <b>Pitch fist</b> | Waving the fist up and down                             | Upwards        |
| <b>Roll hand</b>  | Turning the hand palm up and down with extended fingers | Turn right     |
| <b>Roll fist</b>  | Turning the fist up and down                            | Downwards      |

This seminar will only focus on the discrimination of flexion and extension (the two antagonistic phases) of the index finger.

### Data structure

The experiment originally resulted in two datasets

1. Neurophysiological recordings (#Channels x #Samples) -> ~~rawEcog.mat~~ [rawEcog.pkl](#)
2. Behavioural data (data glove) -> ~~gloveData.mat~~ [gloveResamp.pkl](#)

The ~~'rawEcog.mat'~~ [rawEcog.pkl](#) data has already been transformed and all the data relevant for the course is saved in the file ~~ecogStruct.mat~~. [ecogStruct.pkl](#)